EXP NO : 9
Date : 24/3/26

Determination of Z Parameters of two Port Network

AIM:

To calculate and verify Z parameters of a two port Network using Multisim simulator.

APPARATUS REQUIRED:

S.NO	Components	specification	Quality.
1	Power supply		1
2	Multimeter	---	1
3	Resistors	10KΩ, 10KΩ, 4.7KΩ	
4	Bread board	---	1
5	Connecting wires	---	---
6	PC with multisim simulator	---	---

THEORY:

A two - port network is an electrical circuit or device with two pairs of terminals. This makes possible the isolation of either a complete circuit or part of it. They are import in modeling electronic devices and systemcomponents.

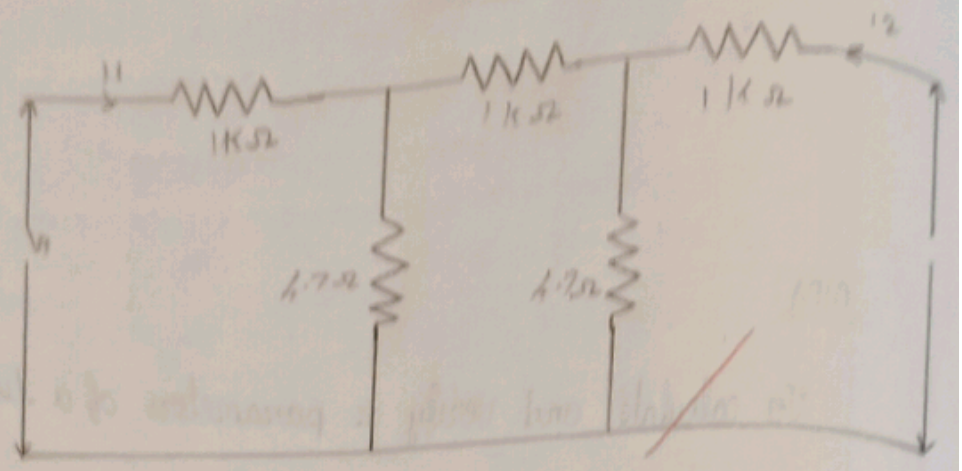
In z parameters of a two - port, the input & output voltage V_1 & V_2 can be expressed in terms of input & output current I_1 & I_2. out of four variables (i.e V_1, V_2, I_1, I_2) V_1 & V_2 are dependent variables whereas I_1 & I_2 are independent variables. Thus-

$$V_1 = Z_{11} I_1 + Z_{12} I_2 \quad — ①$$

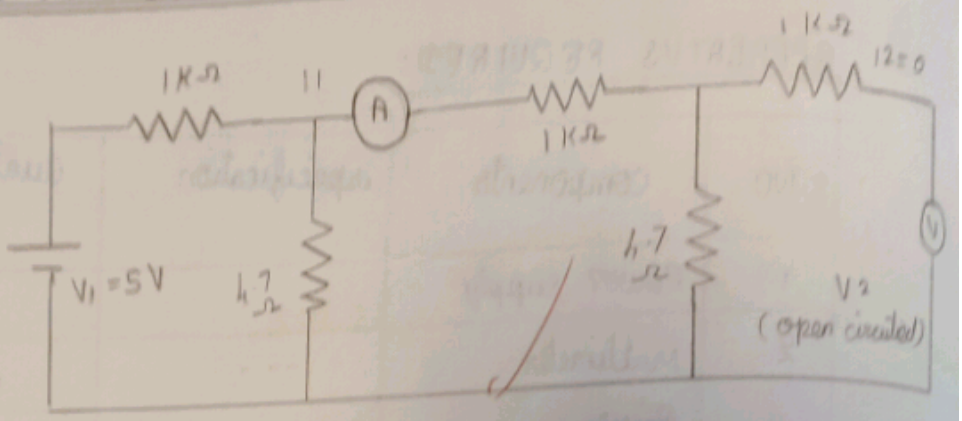
$$V_2 = Z_{21} I_1 + Z_{22} I_2 \quad — ②$$

Here Z_{11} & Z_{22} are the input & output driving point impedance while Z_{12} & Z_{21} are the reverse & reverse & forward tranfer impedances

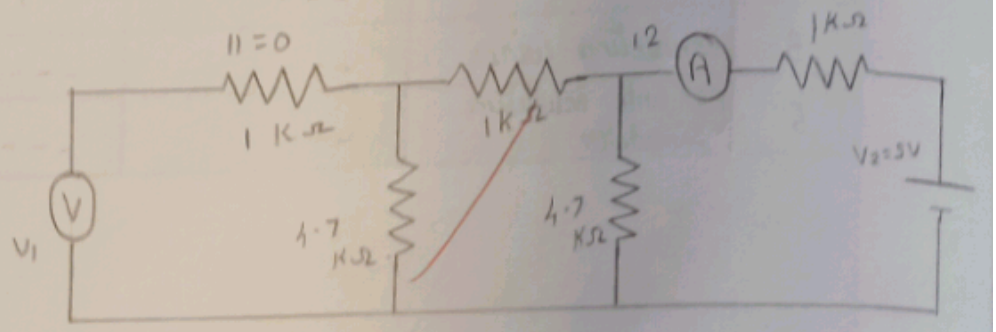
CIRCUIT DIAGRAM:

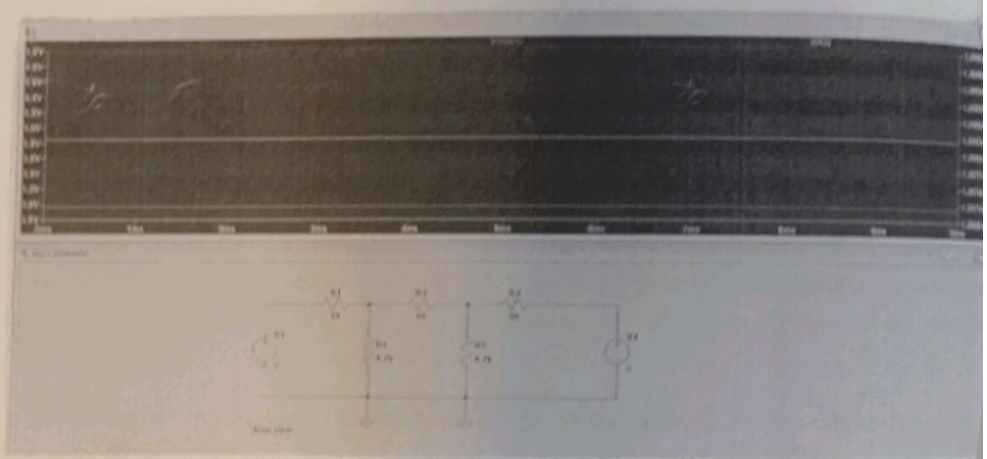


OUTPUT OPEN CIRCUITED:



INPUT OPEN CIRCUITED:





PROCEDURE :

a) Connect the circuit as shown in fig. & switch 'ON' the experimentalboard.

b). First open the O/P terminal & supply 5V to I/P terminal. measure O/P voltage & I/P current.

c). Calculate the value of z parameter using equation (1) & (2)

d). switch 'OFF' the supply after taking thereadings.

OBSERVATION:

S.NO	OUTPUT OPEN CIRCUITED ($I_1 = 0$)			$Z_{21} = V_2/I_1 (\Omega)$	$Z_{11} = V_1/I_1 (\Omega)$
	V_1	V_2	I_1		
1	5 V	2.97 V	1.39 mA	2.136 Ω	3.597 Ω

S.NO.	INPUT OPEN CIRCUITED ($I_1 = 0$)			$Z_{12} = V_1/I_2 (\Omega)$	$Z_{22} = V_2/I_2 (\Omega)$
	V_2	V_1	I_2		
1.	297 V	5V	1.39 mA	2.136 Ω	3.597 Ω

CALCULATION:

Theoretical calculation

To find Z_{11} & Z_{21}

$$V_1 = Z_{11} I_1 + Z_{12} I_2$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2$$

when $I_2 = 0 \rightarrow$ Part 2 open circuit

$$Req_1 = 1 + 4.7 = 5.7$$

$$Req_2 = \frac{5.7 \times 4.7}{5.7 + 4.7}$$

$$= 2.5 + 5$$

$$R = 2.575 + 1 = 3.575 \text{ K}\Omega$$

MARKS SPLIT-UP:

Sl.No	Description	Mark Allotted	Marks obtained
1	Experimental setup / connection	20 marks	20
2	observation / Readings	20 marks	20
3	calculation / Results	20 marks	18
4	Viva	20 marks	18
5	Record	20 marks	18
		Total	94

RESULT:

The Z-parameters of the two port network has been calculated and verified by using multisim simulator.

By ohm's law,

$$V_1 = I_1 R$$

$$\frac{V_1}{I_1} = R$$

$$Z_{11} = \frac{V_1}{I_1} = 3.575\ K\Omega$$

By current divider rule,

$$I_2 = I_1 \times \frac{4.7}{4.7+14.7}$$

$$I_2 = \frac{4.7 I_1}{10.4}$$

$$I_2 = 0.451\ I_1$$

Ohm's law

$$V_2 = I_2 \times 4.7$$

$$V_2 = 4.7 \times I_1 \times 0.451$$

$$Z_{21} = \frac{V_2}{I_1} = 2.1197\ K\Omega$$

$\therefore$ The network is symmetrical

$$Z_{11} = Z_{22} \ \& \ Z_{21} = Z_{12}$$

$$Z = \begin{bmatrix} 3.575 & 2.1197 \\ 2.1197 & 3.575 \end{bmatrix}$$